

Q1 - 25 July - Shift 1

The momentum of an electron revolving in n^{th} orbit is given by : (Symbols have their usual meanings)

(A) $\frac{nh}{2\pi r}$

(B) $\frac{nh}{2r}$

(C) $\frac{nh}{2\pi}$

(D) $\frac{2\pi r}{nh}$

Space for your notes:

Q2 - 25 July - Shift 1

The magnetic moment of an electron (e) revolving in an orbit around nucleus with an orbital angular momentum is given by :

(A) $\vec{\mu}_L = \frac{e\vec{L}}{2m}$

(B) $\vec{\mu}_L = -\frac{e\vec{L}}{2m}$

(C) $\vec{\mu}_L = -\frac{e\vec{L}}{m}$

(D) $\vec{\mu}_L = \frac{2e\vec{L}}{m}$

Space for your notes:

Q3 - 25 July - Shift 2

Hydrogen atom from excited state comes to the ground by emitting a photon of wavelength λ . The value of principal quantum number 'n' of the excited state will be : (R : Rydberg constant)

(A) $\sqrt{\frac{\lambda R}{\lambda - 1}}$

(B) $\sqrt{\frac{\lambda R}{\lambda R - 1}}$

(C) $\sqrt{\frac{\lambda}{\lambda R - 1}}$

(D) $\sqrt{\frac{\lambda R^2}{\lambda R - 1}}$

Space for your notes:

Q4 - 25 July - Shift 2

Questions

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$\frac{x}{x+4}$ is the ratio of energies of photons

produced due to transition of an electron of hydrogen atom from its

(i) third permitted energy level to the second level and

(ii) the highest permitted energy level to the second permitted level.

The value of x will be

Space for your notes:

Q5 - 26 July - Shift 1

In a hydrogen spectrum, λ be the wavelength of first transition line of Lyman series. The

wavelength difference will be " $a\lambda$ " between

the wavelength of 3rd transition line of Paschen

series and that of 2nd transition line of Balmer

Series where $a =$ _____

Space for your notes:

Q6 - 26 July - Shift 2

A nucleus of mass M at rest splits into two parts

having masses $\frac{M'}{3}$ and $\frac{2M'}{3}$ ($M' < M$). The ratio

of de Broglie wavelength of two parts will be :

(A) 1 : 2

(B) 2 : 1

(C) 1 : 1

(D) 2 : 3

Space for your notes:

Q7 - 26 July - Shift 2

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Questions

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Mass numbers of two nuclei are in the ratio of 4:3. Their nuclear densities will be in the ratio of

- (A) 4:3 (B) $\left(\frac{3}{4}\right)^{\frac{1}{3}}$
(C) 1:1 (D) $\left(\frac{4}{3}\right)^{\frac{1}{3}}$

*Space for your notes:***Q8 - 27 July - Shift 1**

An electron (mass m) with an initial velocity $\vec{v} = v_0 \hat{i}$ ($v_0 > 0$) is moving in an electric field $\vec{E} = -E_0 \hat{i}$ ($E_0 > 0$) where E_0 is constant. If at $t = 0$ de Broglie wavelength is $\lambda_0 = \frac{h}{mv_0}$, then its de Broglie wavelength after time t is given by

- (A) λ_0 (B) $\lambda_0 \left(1 + \frac{eE_0 t}{mv_0}\right)$
(C) $\lambda_0 t$ (D) $\frac{\lambda_0}{\left(1 + \frac{eE_0 t}{mv_0}\right)}$

*Space for your notes:***Q9 - 29 July - Shift 1**

Find the ratio of energies of photons produced due to transition of an electron of hydrogen atom from its (i) second permitted energy level to the first level, and (ii) the highest permitted energy level to the first permitted level.

- (A) 3 : 4 (B) 4 : 3
(C) 1 : 4 (D) 4 : 1

*Space for your notes:***Q10 - 29 July - Shift 2**

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Questions

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Read the following statements:

Space for your notes:

(A) Volume of the nucleus is directly proportional to the mass number.

(B) Volume of the nucleus is independent of mass number.

(C) Density of the nucleus is directly proportional to the mass number.

(D) Density of the nucleus is directly proportional to the cube root of the mass number.

(E) Density of the nucleus is independent of the mass number.

Choose the correct option from the following options.

(A) (A) and (D) only. (B) (A) and (E) only.

(C) (B) and (E) only. (D) (A) and (C) only

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Answer Key

Q1 (A)

Q2 (B)

Q3 (B)

Q4 (5)

Q5 (5)

Q6 (C)

Q7 (C)

Q8 (D)

Q9 (A)

Q10 (B)

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Q1 (A)

Angular momentum is integral multiple of $\frac{h}{2\pi}$

$$mvr = \frac{nh}{2\pi}$$

$$\text{So momentum } mv = \frac{nh}{2\pi r}$$

Q2 (B)

Ratio of magnetic moment and angular momentum

$$\frac{\vec{\mu}}{\vec{L}} = \frac{q}{2m}$$

For e^-

$$\vec{\mu} = -\frac{e}{2m} \vec{L}$$

Q3 (B)

$$E_n = \frac{-Rch}{n^2} \quad (1)$$

$$\text{---} \quad n=n$$

$$E_{\text{photon}} = E_n - E_1$$

$$\text{---} \quad n=1$$

$$E_1 = \frac{-Rch}{(1)^2} \quad (1)$$

$$\frac{-Rch}{(n)^2} + \frac{Rch}{1} = \frac{hc}{\lambda}$$

$$\frac{-R}{n^2} + R = \frac{1}{\lambda}$$

$$R - \frac{1}{\lambda} = \frac{R}{n^2}$$

$$\frac{\lambda R - 1}{\lambda} = \frac{R}{n^2}$$

$$n^2 = \frac{\lambda R}{\lambda R - 1} \Rightarrow n = \sqrt{\frac{\lambda R}{\lambda R - 1}}$$

Q4 (5)

$$\frac{13.6 \left(\frac{1}{2^2} - \frac{1}{3^2} \right)}{13.6 \left(\frac{1}{2^2} - 0 \right)} = \frac{x}{x+4}; \quad \frac{\frac{1}{4} - \frac{1}{9}}{\frac{1}{4}} = \frac{x}{x+4}$$

$$\frac{5}{9} = \frac{x}{x+4}$$

$$5x + 20 = 9x$$

$$4x = 20$$

$$x = 5$$

Q5 (5)

For first line of Lyman

$$\frac{1}{\lambda} = R \left(1 - \frac{1}{4} \right) = R \left(\frac{3}{4} \right)$$

$$\Rightarrow \lambda = \frac{4}{3R} \dots (1)$$

3rd line (Paschen)

$$\frac{1}{\lambda_3} = R \left(\frac{1}{3^2} - \frac{1}{6^2} \right) = \frac{R}{9} \times \frac{3}{4}$$

2nd line (Balmer)

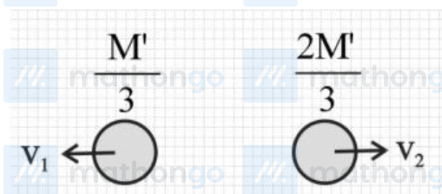
$$\frac{1}{\lambda_2} = R \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \frac{R}{4} \times \frac{3}{4}$$

$$\text{Thus } a\lambda = \lambda_3 - \lambda_2 = \frac{12}{R} - \frac{16}{3R} = \frac{20}{3R} \text{ putting (1)}$$

$$a \left(\frac{4}{3R} \right) = \frac{20}{3R} \Rightarrow a = 5$$

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Q6 (C)



$$|\vec{P}_1| = |\vec{P}_2|$$

Here $\vec{P}$ is momentum

$$\text{So } \lambda = \frac{h}{p}$$

Hence both will have same de broglie wavelength.

Q7 (C)

$$\text{Radius of nucleus } R = R_0 A^{\frac{1}{3}}$$

$$\text{Density of nucleus} = \frac{\text{Mass of nucleus}}{\text{volume of nucleus}}$$

$$\rho = \frac{m \times A}{\frac{4}{3} \pi R^3} \quad \text{Where } m : \text{mass of proton or neutron}$$

$$\rho = \frac{m \times A}{\frac{4}{3} \pi R_0^3 A}$$

$$\rho \propto A^0$$

Hence density of nucleus is independent of mass number

Q8 (D)

$$\ominus \rightarrow V_0$$

$$E = -E_0 \hat{i}$$

$$\lambda_0 = \frac{h}{mv_0}$$

$$v = v_0 + \frac{eE_0 t}{m}$$

$$\lambda = \frac{h}{mv} = \frac{h}{m \left(v_0 + \frac{eE_0 t}{m} \right)}$$

$$\lambda' = \frac{h}{mv_0 \left(1 + \frac{eE_0 t}{mv_0} \right)}$$

$$\lambda' = \frac{\lambda_0}{\left(1 + \frac{eE_0 t}{mv_0} \right)}$$

Q9 (A)

$$E_n = \frac{-13.6}{n^2} \text{ eV}$$

$$\Rightarrow \frac{E_2 - E_1}{E_\infty - E_1} = \frac{13.6 \left(1 - \frac{1}{4} \right)}{13.6} = \frac{3}{4}$$

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Q10 (B)

$$R \propto A^{1/3}$$

$$V = \frac{4}{3}\pi R^3 \propto A$$

$$\text{Mass} \propto A$$

So density is independent of A.

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